AZ-305 Agenda

Module 01 Design a governance solution

Module 02 Design a compute solution

Module 03 Design a non-relational data storage solution

Module 04 Design a data storage solution for relational data

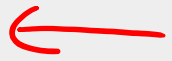
Module 05 Design a data integration solution

Module 06 Design an application architecture solution

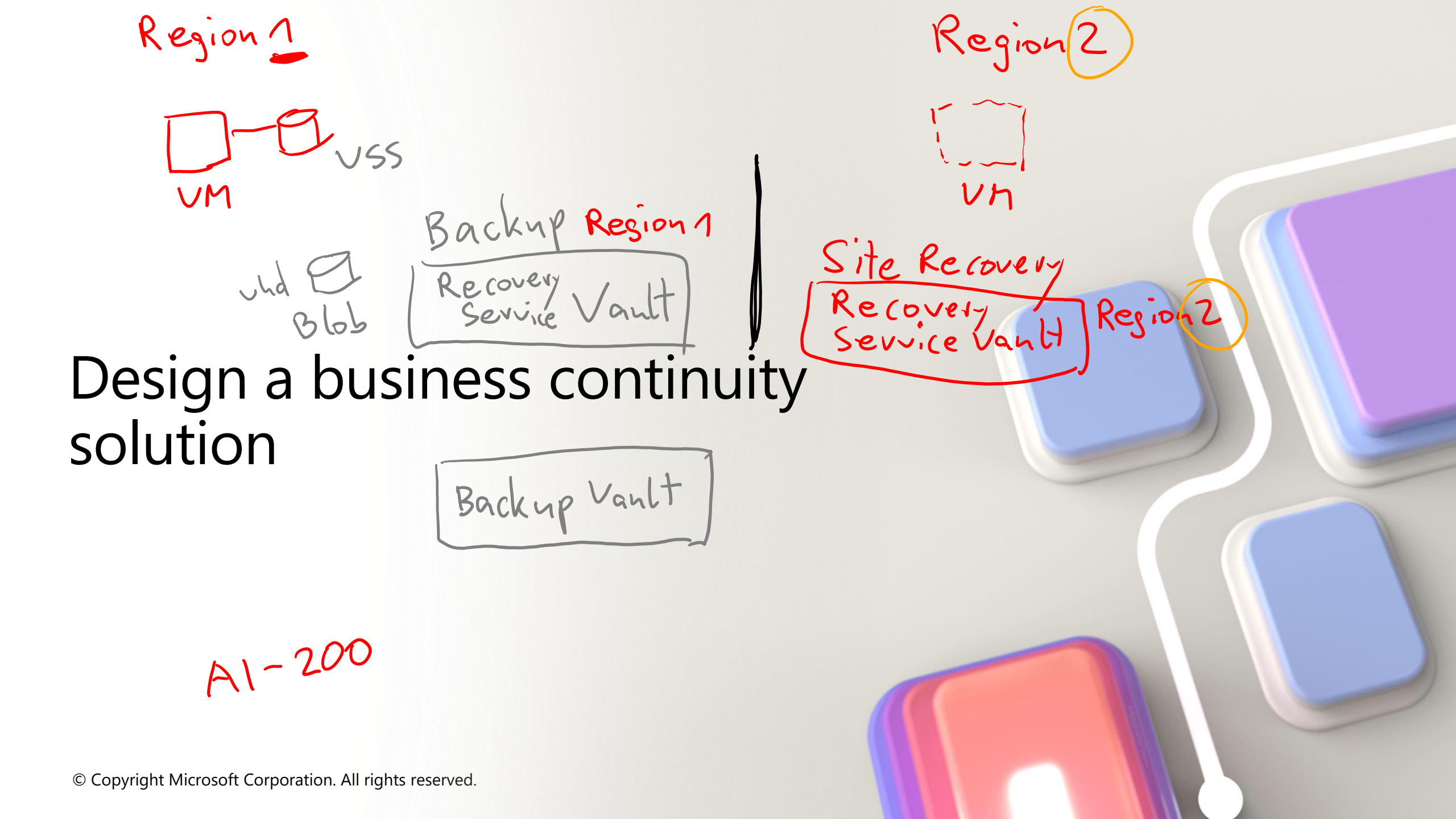
Module 07 Design Authentication and Authorization Solutions

Module 08 Design a solution to log and monitor Azure resources

Module 09 Design a network infrastructure solution

Module 10 Design a business continuity solution 

Module 11 Design a migration solution



Design a business continuity solution

AI-200

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Learning Objectives

- Design for backup and recovery
- Design for Azure Backup
- Design for Azure blob backup and recovery
- Design for Azure Files backup and recovery
- Design for Azure virtual machine backup and recovery
- Design for Azure SQL backup and recovery
- Design for Azure Site Recovery
- Learning recap

AZ-305: Design Business Continuity Solutions (15-20%)

Design a Solution for Backup and Disaster Recovery

- Recommend a recovery solution for Azure and hybrid workloads that meets recovery objectives
- Recommend a backup and recovery solution for compute
- Recommend a backup and recovery solution for databases
- Recommend a backup and recovery solution for unstructured data

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Design for backup and recovery



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Plan for backup and recovery

Identify your business needs and create a plan to address those needs

- What are your workloads and their usage?
- What are the usage patterns for your workloads?
- What are the availability metrics (MTTR and MTBF)?
- What are the recovery metrics (RTO and RPO)?
- What are the workload availability targets?
- What are your SLAs?

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Design for Azure Backup

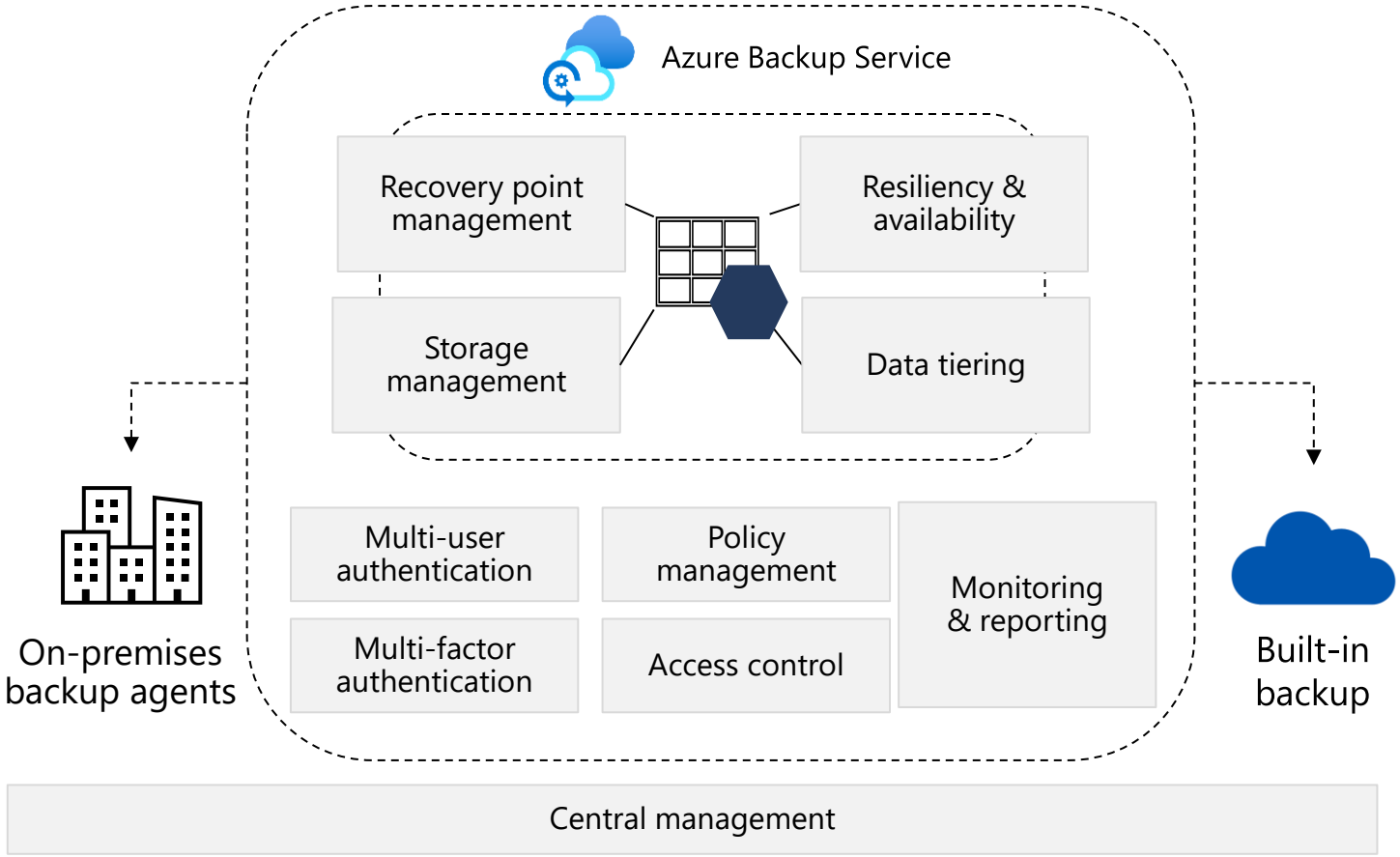


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When to use Azure Backup

Azure Backup is a full-service backup and recovery solution.

- Unlimited scaling with high availability and unlimited data transfer
- Automatic replication of locally redundant storage and geo-redundant storage using a pay-as-you-use model
- Application-consistent backups with secure transmission and storage of your data in Azure
- No limits on the length of time you can keep the backup data



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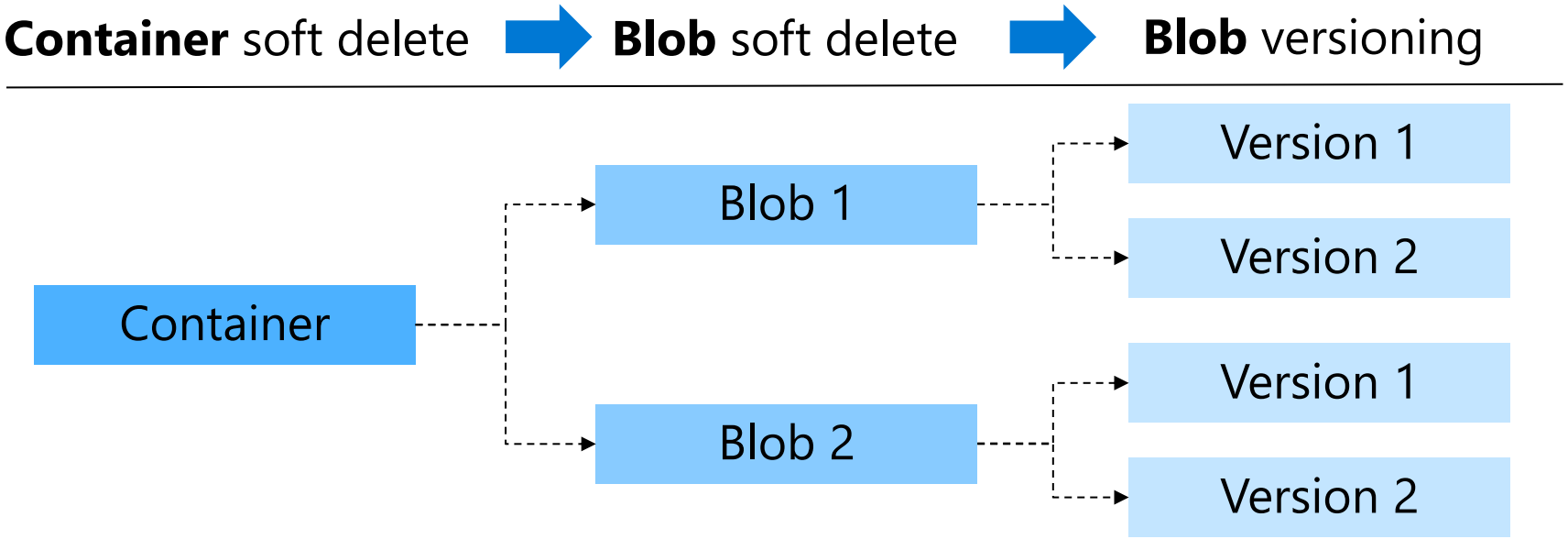
Design for blob backup and recovery



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Considerations for soft delete

Consider soft delete with recovery times from 1 to 365 days



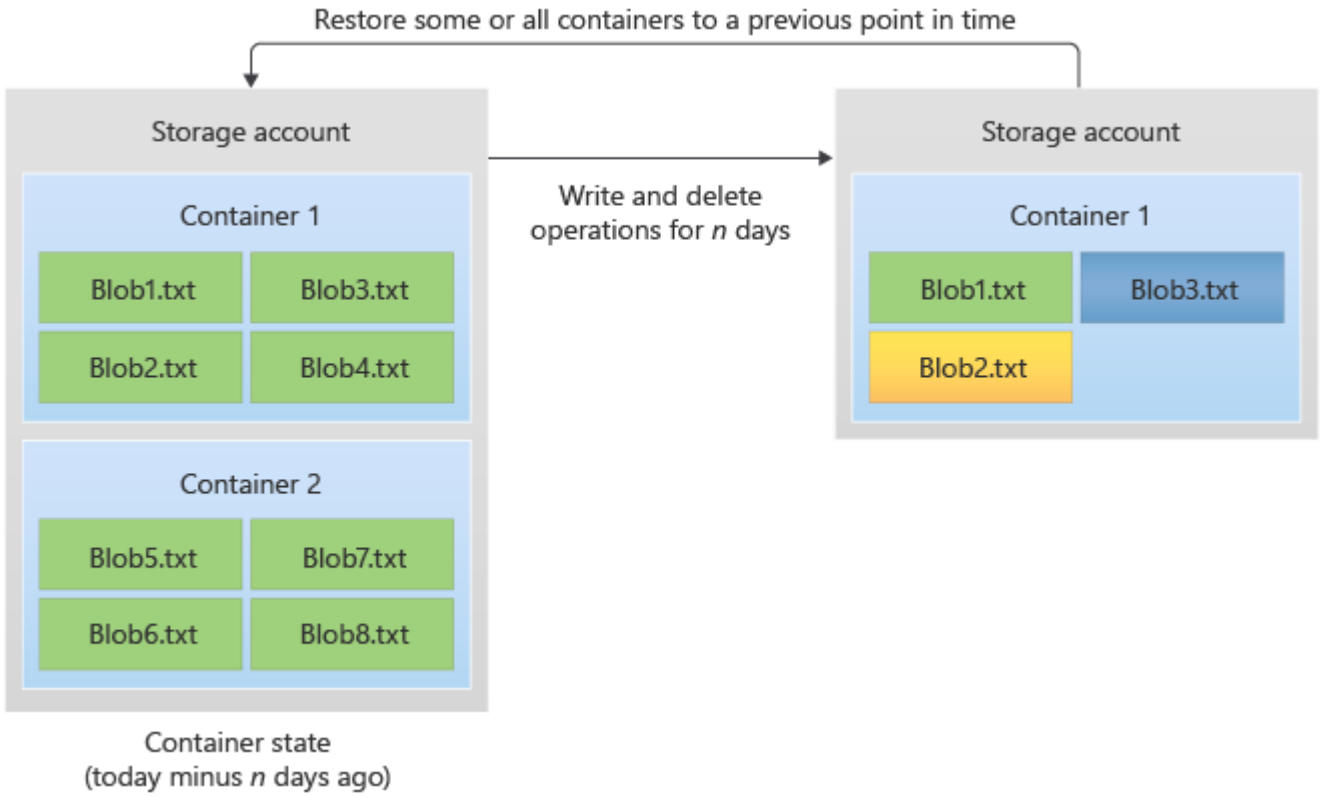
- Maintains the deleted data in the system for a specified retention period
- Soft delete protects blobs, snapshot, containers, or versions from accidental deletes or overwrites

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Considerations for point-in-time restore

Consider point-in-time restore for block blobs

- Useful in scenarios where a user or application accidentally deletes data or where an application error corrupts data
- Use policy to specify the retention period



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Design for Azure Files backup and recovery

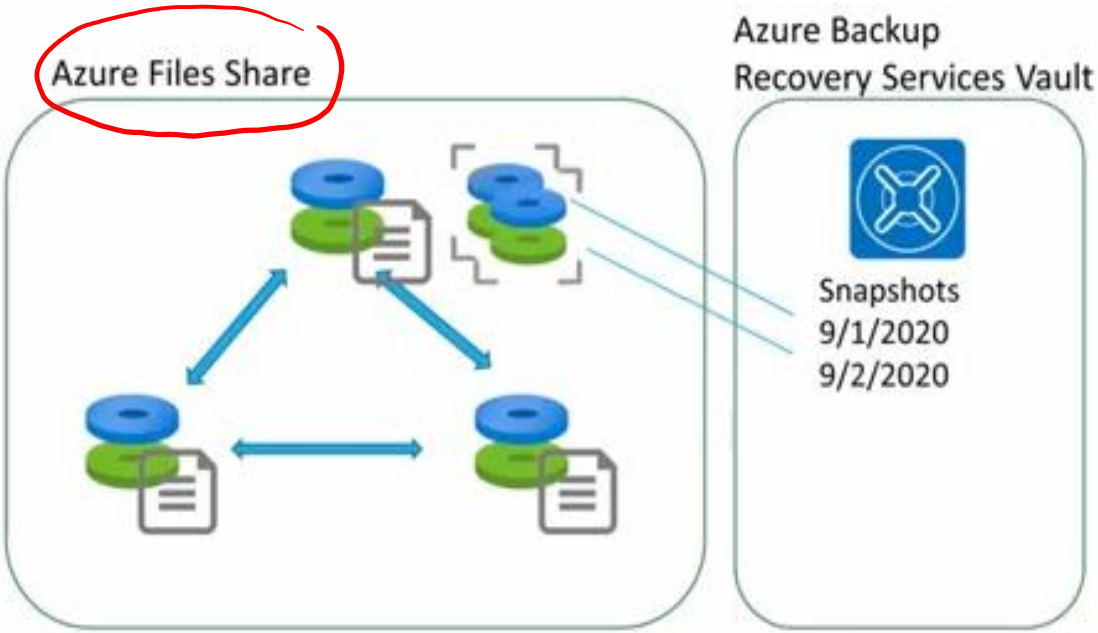


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Considerations for Azure Files backup and recovery

Consider snapshots for both blobs and Azure Files

- Organize file shares with backup in mind
- Snapshots can be on-demand or scheduled using Azure Backup and backup policies.
- Snapshots are at the file share root – retrieval is at the file
- Use snapshots to cover the time between daily backups
- Use instant restore – consider self service restore
- Snapshots are incremental - snapshot before code deployments.



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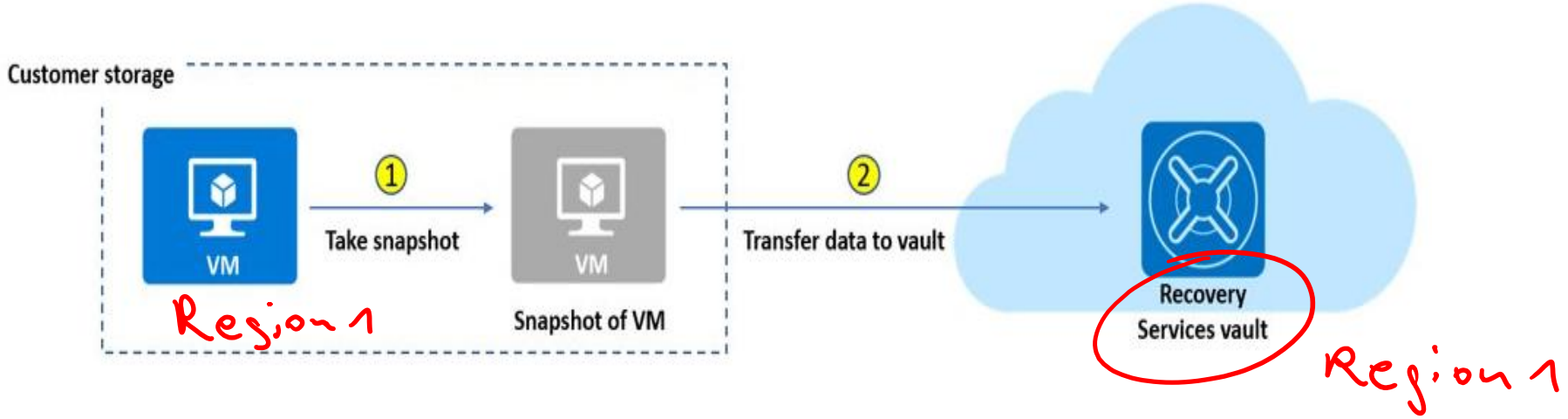
Design for virtual machine backup and recovery



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Considerations for Azure virtual machines

Guard against unintended destruction of the data on your VMs.



- Group VMs into customized backup policies
- Combine short-term (daily), long-term (weekly), and on-demand backups
- Identify needs for app, crash, and file backups – practice the restore
- Consider Cross Region Restore (CRR) for VMs in the paired region
- Periodically review your policies – add monitor and alert

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Design for Azure SQL backup and recovery



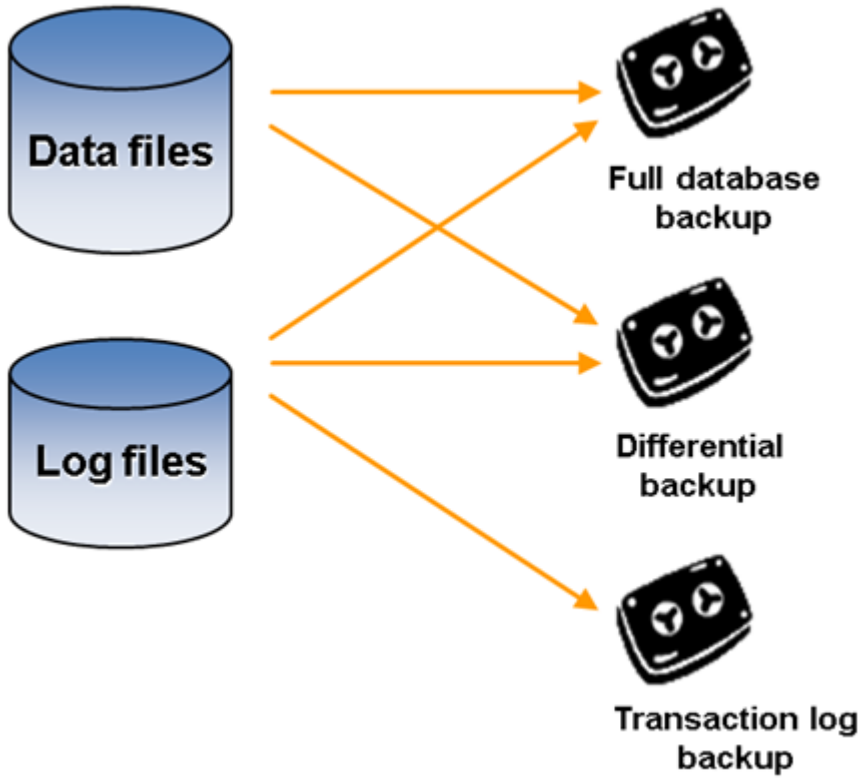
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How Azure SQL backup works

SQL Database and SQL Managed Instances automatically backup.

For fully consistent backups, the following are taken:

- Full backups once a week
- Differential backups every 12-24 hours
- Transactional log backups every 5-10 minutes



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Considerations for Azure SQL backup

Restore in the retention period or use a long-term retention policy

- Restore an existing database to a point in time in the past within the retention period
- Restore a deleted database to the time of deletion or to any point in time within the retention period
- Restore a database to another geographic region
- Restore a database from a specific long-term backup of a single database or pooled database
- Long term retention uses read-access geo-redundant storage (RA-GRS)



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Design for Azure Site Recovery

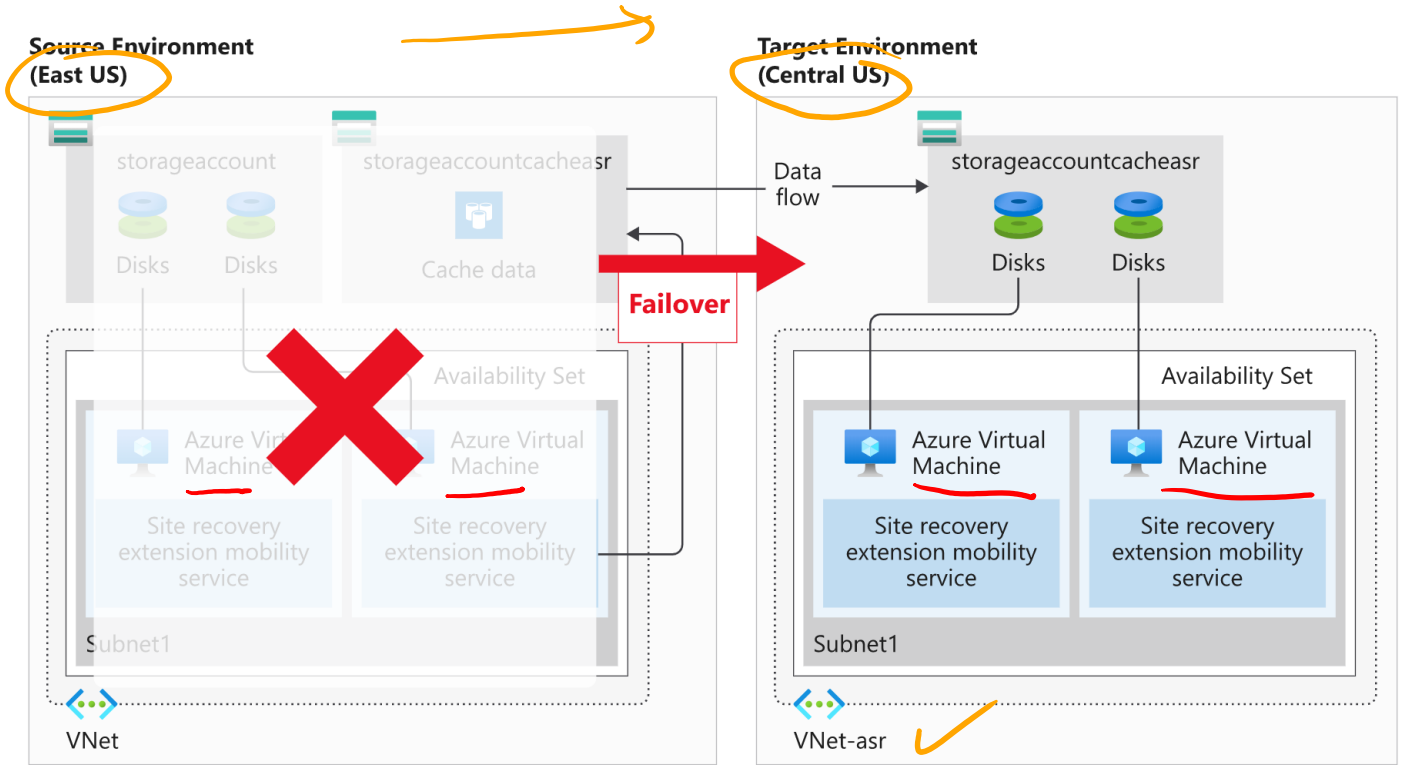


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When to use Azure Site Recovery

Failover for Azure, on-premises, other cloud provider resources

- Perform disaster recovery and validate the replication strategy
- Migrate on-premises VMs and physical servers to Azure
- Replicate virtual machines between regions
- Define retention history and frequency of snapshots



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Combine Azure Site Recovery with Azure Backup

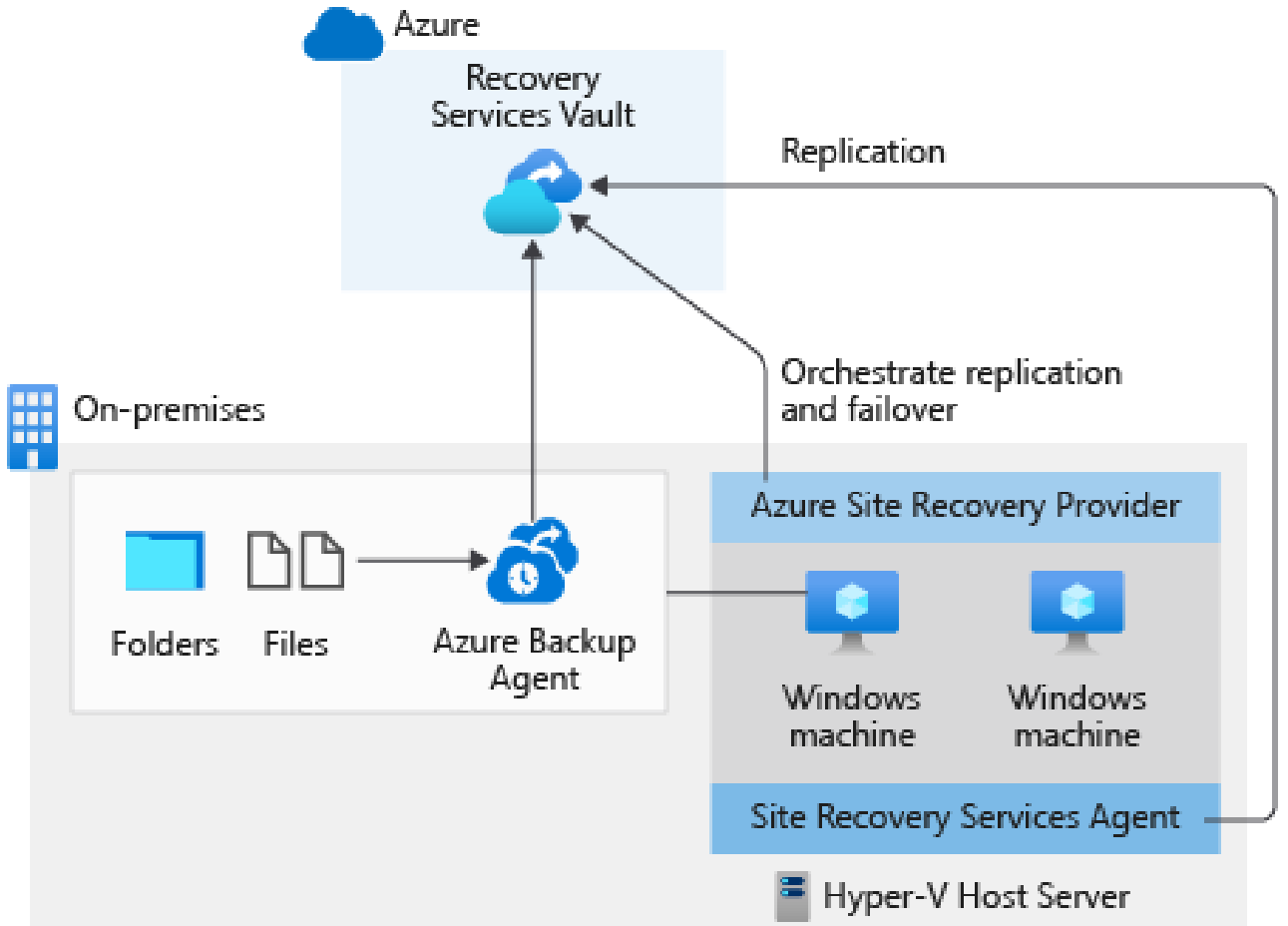
Combine ASR with Azure Backup

Requirement

- Backup all the files and folders in this virtual machine to Azure.
- Protect any workloads running on the virtual machine and keep running them even if the virtual machine fails.

Azure Backup

- Azure Backup periodically backs up the files and folders on the Windows machine to Azure.
- This process ensures they are secure and retrievable even if the whole on-premises environment stops functioning.



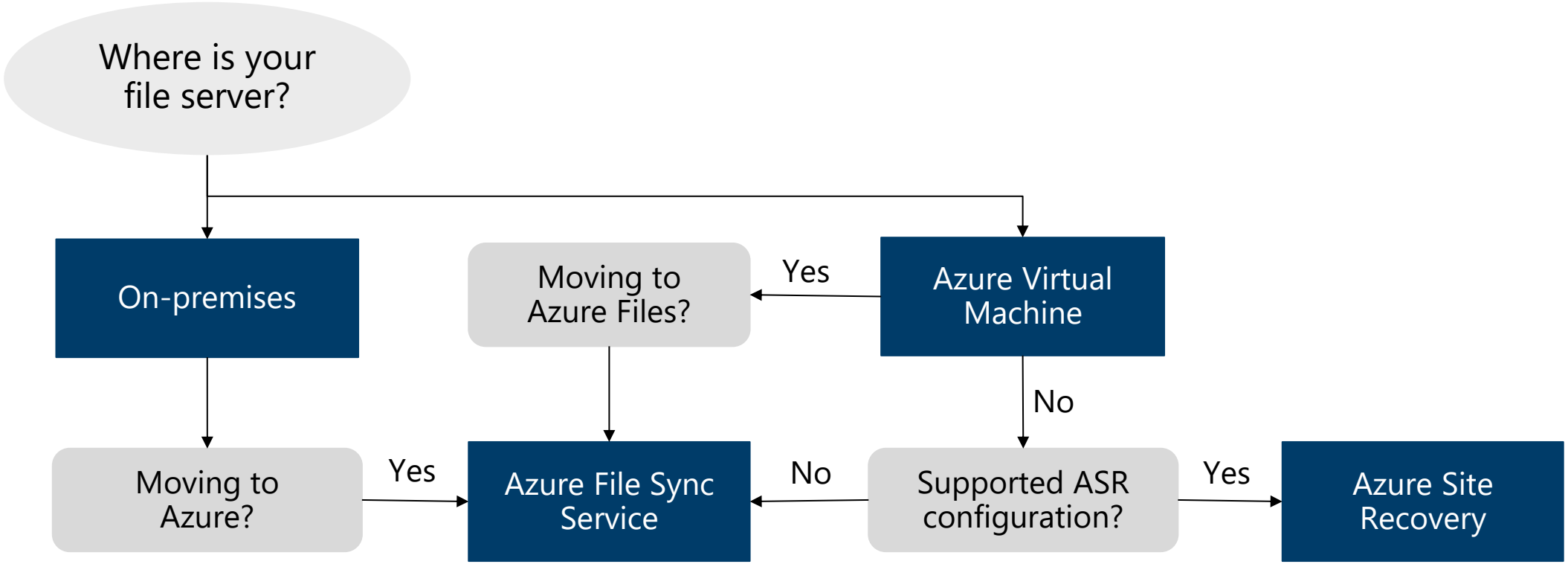
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Review



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Review file server backup and recovery options



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Recommend a disaster recovery method (activity)



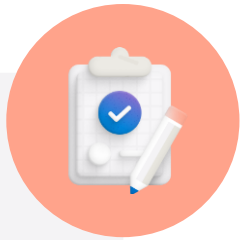
- Azure Backup
- Azure Site Recovery
- VM Snapshot
- MABS

You need to back up on-premises machines and workloads	You need to back up your managed disks at any point in time
You have Azure VMs running production workloads	You need a read-only full copy of a managed disk
You need application-consistent backups for Linux virtual machines	You need to cover disaster scenarios like an entire regional outage

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Learning Recap – Business continuity solutions

Check your knowledge questions and review



Reference modules

- Protect your virtual machines by using Azure Backup
- Disaster recovery and backup

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Instructor resources (hidden)



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End of presentation



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